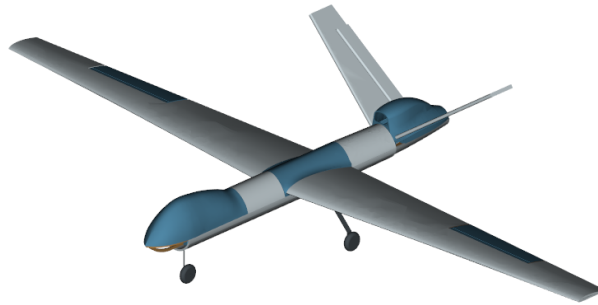


01 / MQ9 airframe

Manufacturing and assembly guide / 2 metre RC model / Revision 1



For the **Manufacturing Team**: fabricate and assemble the body. The **Electronics Team** installs and configures electronics using its separate guide.

Start here

1. Read this guide and the part registers. Print the fit coupons first.
2. Cut plywood and make the printed/metal parts. Mark every part ID.
3. Dry-assemble the complete skeleton and removable wings.
4. Form and fit one representative foam panel, then finish the skins.
5. Deliver the body with access open for electronics, plus the completed inspection record.

Before flight: complete the fit, material, structural, propulsion and balance checks. Digital checks do not establish flight performance.

02 / Files, tools and stock

Folder	Use
01_Plywood	Individual CUT profiles, quantity register and revised positioned STEP parts.
02_Foam	Foam patterns, forming instructions and panel locations.
03_Print	Only selected aircraft print STLs, matching STEP and settings.
04_Hardware	Tube schedule, metal forming/drilling references, motor drawing and coupon.
05_Assembly	Revised assembly STEP and labelled reference illustrations.
Build Guide	Instructions, consolidated manual, part atlases and overviews.
Print_Templates	Actual-size A4 cutting and tooling templates. Keep these separate from the reading manual.

Machine work versus bench work

Laser / CNC: plywood CUT layer only.

Printer: selected aircraft STLs.

Metalwork: form and drill landing gear.

Knife: foam skins. Laser-cut XPS only with workshop approval for this material and process.

Bench work: dry-fit, bevel, form, glue and align parts; install fasteners and linkages; check moving clearances.

Stock and setup

Use aircraft plywood at the specified thickness, including 2.5 mm nose doublers. Skins use 5 mm FliteBoard Pro, 1000 x 600 mm. Use the specified carbon tubes; 5 mm solid rods are not substitutes.

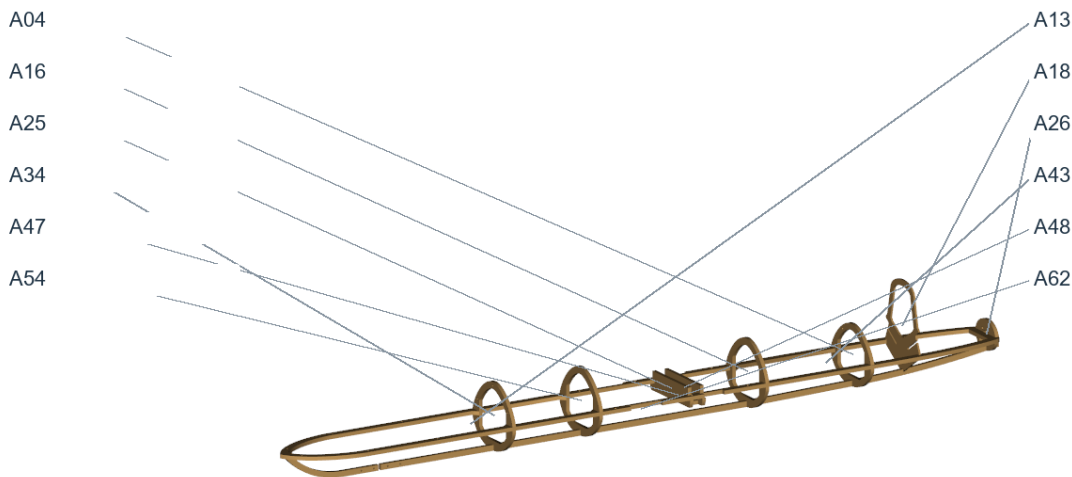
Sheet layouts use 1220 x 2440 mm stock. The shop may rearrange parts at 100% scale. Preserve quantities and grain along spars, spine and doublers.

03 / Parts and buying list

Item	Quantity / specification
Plywood	31 cut pieces from 22 profiles: 24 nominal 5 mm, 4 at 3 mm, 3 at 2.5 mm. Two outer ribs are 4.990 mm CAD thickness; finish 5 mm stock to fit.
Foam	5 mm FliteBoard Pro. Follow per-pattern quantities; moving tail controls use 2 layers each. Ten supplied sheets are starting stock, not ten installed sheets.
Printed aircraft parts	11 individual STLs. One fit coupon is additional and is not installed.
Carbon	Front 8/6 mm joiner 406 mm x 1; rear 8/6 mm 390 mm x 1; matched 10 mm OD/8.2 mm nominal ID sleeves 199.35 mm x 2.
Wheels and gear	3 lightweight 50.8 mm wheels, 3 mm bores; formed main gear and 3 mm spring-wire nose gear; steering arm and shaft collars.
Wing root retention	6 M2 bolts, start 20 mm long; 12 washers and 6 locking nuts. Verify final stack.
Nose bypass	2 M2x 40-45 inner bolts +2 M2x 16 outer bolts; 8 washers, 4 locking nuts. Select actual length from stack.
Main gear fastening	2 M3x 16 through-bolts, large washers, 2 locking nuts; 2 M3 axle bolts, nominal 25 mm plus spacers/nuts.
Motor fastening	4 M3 screws, matching washers; determine screw length from 5 mm ply and actual safe motor thread depth.
Nose bearing	4 M3 bolts through bearing/mount, washers and locking nuts; start 16-20 mm and verify stack/clearance.
Controls	5 servos: 2 ailerons, 2 tail, 1 nose steering. 4 control horns, 1 steering arm, 5 pushrods with 10 clevises/ball links; servo screws.
Hinge pins	2 smooth 3 mm rods; 285 mm initial length, trim after fitting retainers.
Adhesives / covering	Wood-compatible epoxy/wood adhesive; XPS-compatible foam adhesive; reinforced tape and paper seam straps. Qualify on offcuts.

Fastener lengths are starting sizes. Select final lengths from the actual washer, nut and part stack. Buy spare washers, locking nuts, cable ties and servo extensions.

04 / Fuselage skeleton



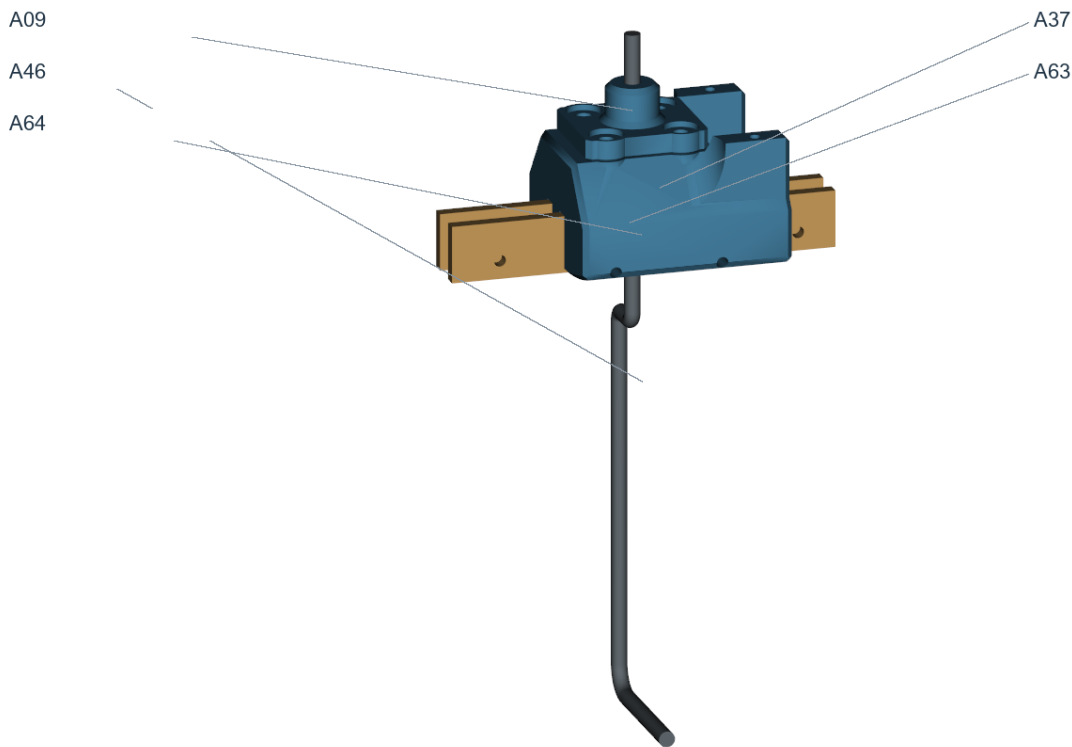
Assembly axes: X = span, Y = nose to tail, Z = up. Nose tip: Y = -439.65 mm. Use the revised assembly STEP for part positions.

1. Lay A62F and A62R on a flat jig along the centreline; first make the nose bypass on page 5. Do not glue them end-to-end across the shaft gap.
2. Fit forward longitudinal member A13 and aft member A43, then former A34 atY=-130, A54 atY=-5, A16 atY=233.65 and A04 atY=384.65. Slots determine final seating.
3. Fit the main-gear box A25/A48 with bottom plate A47. Fit tail bulkhead A18 and revised motor mount A26.
4. Check squareness from the assembly axes, equal left/right former extents, and that all tabs seat without forcing. Test fit the printed nose, centre and tail fairings before bonding.

Bond after dry-fit. Add modest fillets at gear, wing-root and motor joints. Keep electronics access open. Adjust the cutting process using a coupon; never scale a DXF to correct slot fit.

Spine: install A62F/R with the bolted side cheeks. There is no separate A40 piece.

05 / Nose gear bypass



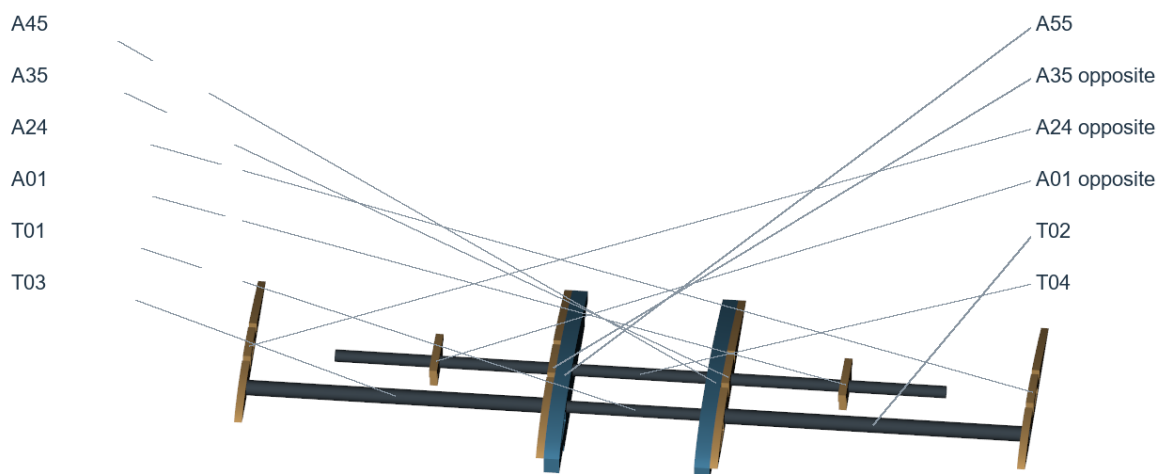
Nose joint: two 2.5 mm plywood cheeks, A63/A64, bridge the shaft gap. Each is 72 x 10.7 mm, with grain along its length. Fit them into the A37 pockets.

1. Place A62F/R collinear with the 3.4 mm shaft gap centred on $Y = -293.65$.
2. Fit one cheek on each side of the 5 mm spine. Dry-fit them into the revised A37 pockets.
3. Install four transverse M2 bolts at $Y = -320, -302, -282, -264$ mm, $Z = -37.964$ mm. Inner bolts pass through the mount; outer bolts clamp the wood stack. Use washers and locking nuts.
4. Fit A09 and the extended 3 mm wire. Rotate the shaft fully. Bond wood interfaces only after the assembly stays straight and turns freely; keep glue out of the shaft passage.
5. Tighten enough to seat the stack without crushing plywood/plastic. Recheck after cure.

Fit the 3 mm wire through A09 into its steering arm/collar. The supplied shaft extends to $Z = +1$ mm. Trim only after full clamp engagement. Retain mechanically; do not rely on glue.

Check: support the body and turn the wheel both ways. Check mount, cheeks and skin together for rubbing, cracks and loose bolts. Recheck under the intended gear deflection.

06 / Carbon joiners and retention



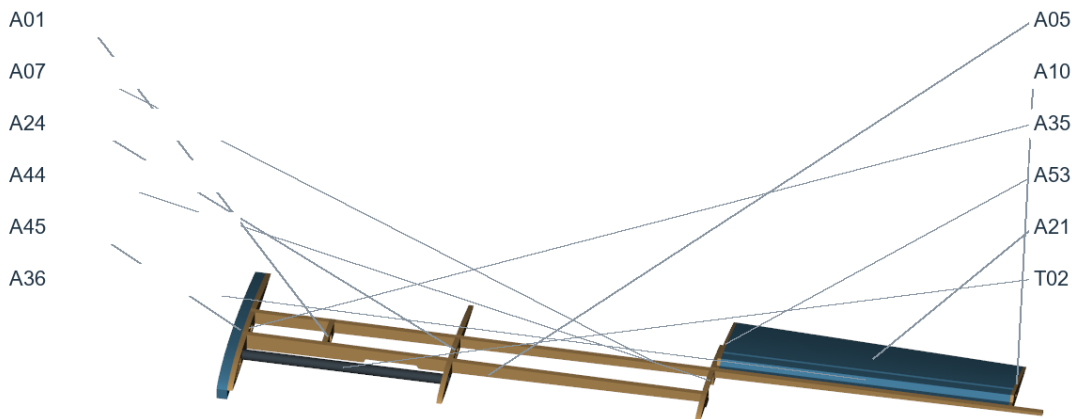
Part	Section / cut length	Position and purpose
T01	8 OD /6 ID; 406 mm	CentreX=-203..203. Front joiner axisY60.454, Z15.632.
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	RightX53..252.35; leftmirror. Bond sleeves into wing root/middle ribs.
T04	8 OD /6 ID; 390 mm	CentreX=-195..195. Rear axisY100.500, Z15.070.

Use a matched sliding tube pair. Nominal 8.2 mm sleeve ID gives 0.2 mm diametral clearance on the 8 mm joiner. Check actual stock. Do not force tubes or sand away structural wall thickness. Wood bores: 10.2 mm for sleeves; 8.2 mm for root/rear supports.

1. Centre both joiners and dry-fit both wings together.
2. Check about 150 mm front-joiner insertion per wing.
3. Bond sleeves only; keep glue off the sliding joiner.
4. Fit three M2 retainers per wing and mark tube insertion positions.
5. Check twist, spanwise retention and hinge clearance.

Use only the four tubes in the schedule. The 5 mm solid carbon rods are spare material.

07 / Wing frames and ailerons



Build a right wing and its left-hand mirror. Follow register quantities and assembled STEP positions.

1. Fit inner rib A35, middle rib A24 and outer rib A44 to front spar A05 and rear spar A07. Fit rear joiner support A01 and the front sleeve. Match each rib slot, not a guessed spacing.

2. On a flat jig, check leading/trailing edges and equal incidence before bonding. Finish A24/A44 edges to the marked noncut sanding guide where needed; preserve revised bore centres.

3. Install the aileron end plates A53/A10, hinge rail A36 and aileron A21 on the right; use A23/A11 on the left. Dry-fit smooth 3 mm pins and retain them mechanically, leaving about 1 mm axial freedom.

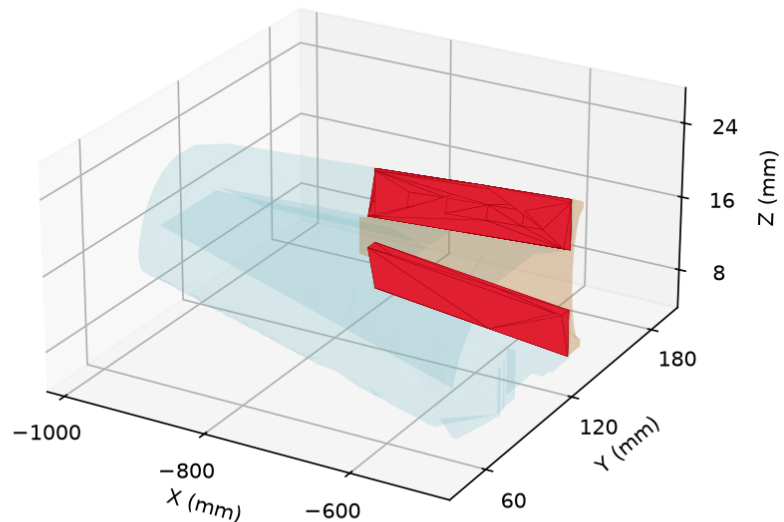
4. Install each servo with its manufacturer screws and a serviceable mounting pad. Transfer actual servo mounting holes after confirming its size; the CAD servo is a reference envelope.

5. Fit the control horn and neutral pushrod before closing the skin. Check travel with no scraping, stiffness or exposed sharp pin end.

Check: ailerons rotate freely on retained pins. Keep glue out of hinges. Finish a tight bore to the actual pin; do not scale the STL.

08 / Foam skins and seams

Wing hinge inner core | red = CAD interference to relieve



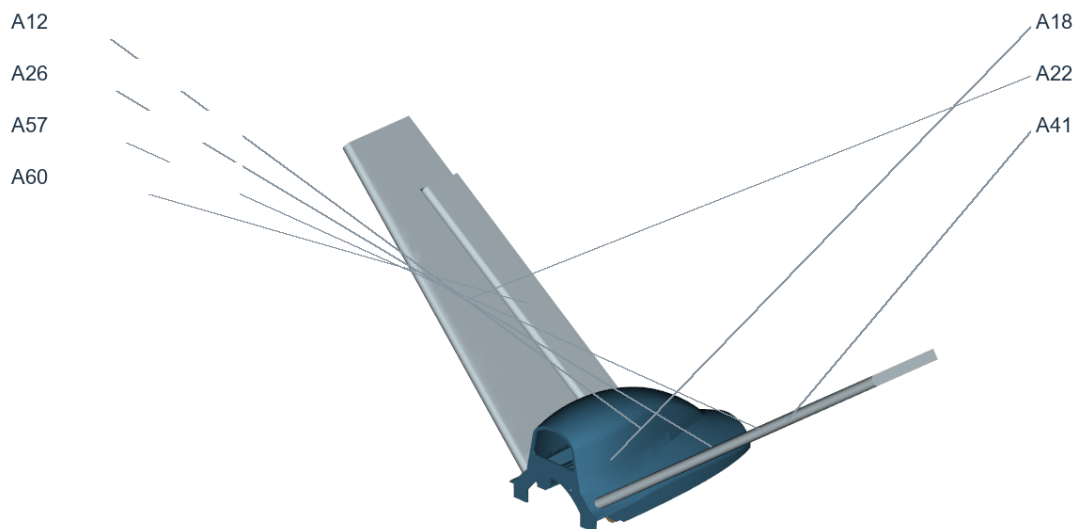
Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

Use F-series patterns. Print at 100%, measure the calibration rectangle, and join matching marks with the indicated 10 mm overlap.

1. Mark each outside face and part ID. Knife-cut the finished skin edge. Moving-tail blanks have 5 mm sacrificial shaping stock. F23/F24 include a 1 mm PAPER-ONLY lower seam lip: remove its core and trim/fold it against the mating lip as shown in the foam guide. Other skin CUT lines have zero added lap/trim.
2. Peel inside paper only in the curved forming zones. Keep outside paper intact as the pattern reference. Gradually relieve/compress inside foam and dry-form over the real frame; do not stretch a short paper edge into place.
3. Seat the wing from its leading-edge region toward the open/trailing seam. Shave local inner foam for the printed hinge rail and rib contact patches. Do not cut away the outside paper to hide an inner interference.
4. Bevel inside seam edges to their actual meeting angle. Butt outside paper edges and add 15 mm internal paper straps, about 7.5 mm per side. Reinforce exposed seams with suitable light tape.
5. Dry-fit an entire wing and its moving surface before gluing progressively along the frame. Hold alignment while adhesive cures.

Check a forming coupon first. Peeled board loses stiffness and may warp. Hold both sides evenly while bonding. Use XPS-compatible adhesive and reinforce inner seams. Ordinary CA must not touch exposed XPS unless explicitly approved for it.

09 / Fuselage covering and tail



Dry-fit A59 nose, A06 centre fairing and A12 tail fairing. Keep a removable access panel for the battery, FC and wiring.

Follow the foam assembly map. Fit lower forward, middle and upper panels, then rear-underside bands in numbered order. Support butt seams inside. Relieve only inner foam where it contacts the spine or longitudinal members.

Trace installed gear before cutting exits. Start with 0.5 mm clearance around static legs, then check steering and deflection. Reinforce the paper edge. Gear loads must pass into the structure, not the foam.

Fixed tails: form skins with the removable collars; test A12 slot engagement.

Moving tails: laminate two 5 mm blanks. Transfer the finished outline, remove the 5 mm sacrificial perimeter and shape long edges to R5.

Tape-hinge on the indicated hinge line. Fit two servos, horns and pushrods with adjustment access. Check both tails for symmetry and free travel before glue cures. Keep straps and glue out of moving gaps.

10 / Printing and metalwork

Print group	Starting process settings, then qualify on a coupon
A06/A12/A59 shells	LW-PLA preferred for mass. If using available PLA+, start 0.2 mm layers, 2 perimeters and 3-5% infill; record slicer mass and weigh the finished part. Support internal overhangs without blocking sockets.
A11/A21 ailerons; A23/A36 rails	Start 3 perimeters, 10% infill; orient long axis on the bed with supports kept out of hinge bores. Use smooth pin sample and full travel check.
A09/A37/A45/A55 functional parts	Prefer a shop material/process suited to load and temperature; available PLA+ requires inspection/testing. Start 4 perimeters, 30% infill. Orient so tube/bolt loads do not simply split layer interfaces. Do not approve from infill percentage alone.

Print in millimetres at 100%. The shop selects orientation and supports. Some parts exceed 220 mm beds: use a larger printer. A smaller printer needs a designed joint, not an arbitrary split or rescale.

Check: clear every passage, measure dimensions, inspect layer bonds and reject cracked or warped brackets. Weigh printed parts for the Electronics Team.

Landing gear

Main gear A02: 12.7 mm wide x 3.175 mm metal strip. The DXF is a formed front view, not a flat blank. The shop selects bendable alloy/temper, bend radii and allowances while preserving attachment and axle positions.

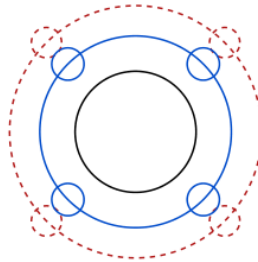
Fit A02 through its two 3.2 mm holes to A47 using M3 bolts, washers and locking nuts. Fit three 50.8 mm wheels. Form the 3 mm nose wire to its STEP reference. Check gear deflection at completed aircraft mass.

11 / Motor mount and team transfer

A26 motor mount - declared drilling correction

Source: 4x3mm on 25mm PCD. Revised: 4x3.2mm on 19mm PCD.

Physical Emax motor check required before drilling; drawing below at 2:1.



Red dashed: obsolete 25mm PCD - DO NOT DRILL

Blue: 19mm PCD - four 3.2mm holes

Central 12mm hole unchanged. Minimum radial wood web 1.9mm; use washers.

A26 motor mount: four 3.2 mm holes on a 19 mm pitch circle; adjacent hole centres 13.435 mm. Preserve the 12 mm centre opening.

Check the actual motor against the template before production. Use washers and measured screw engagement; screws must not reach windings. Keep wires clear of the motor bell.

Propulsion check required. Emax recommends 6-7-inch props. The 9-inch prop / 40 A ESC combination needs measured current, thrust and temperature checks. Keep the propeller off during assembly and calibration.

Leave these accessible

Keep access to battery straps, FC, ESC cooling, receiver, GPS, pitot tubing, wing connectors, horns, steering and gear fasteners. Fit wire pull strings before closing skins.

Transfer the body to the **Electronics Team** with recorded masses, fit notes and labelled loose hardware. Use the build record for outstanding checks.

12 / Checks before the body is signed off

Check	Builder records
Scale and coupons	100 x 20 mm rectangle measured; actual ply/foam thickness; kerf/slot result; printer coupon result.
Structure	All 31 wood pieces accounted for; no source A40 duplicate; bypass bolts installed; no split grain/cracks; aligned centreline.
Wing attachment	Both wings fit simultaneously; matched sleeve fit; centred tubes; six M2 retainers; no spanwise escape.
Controls	Two ailerons, two tail controls and nose steering free; hinge pins retained; horns secured; no glue in moving gaps.
Skins	Full surface dry-fit; seams meet without forced stretching; core relief/gear exits complete; outside paper sound; no wing twist.
Gear and motor	Positive gear fasteners; free wheels; free steering; motor pattern/screw engagement checked; ground clearance examined.
Mass and access	Total body and individual print masses recorded; battery/FC access open; wires can be serviced.
Outstanding evidence	Material forming, adhesive/print strength, gear loading, propulsion and flight checks must be recorded physically.

Balance reference for the Electronics Team

Provisional mark: 522 mm aft of the nose, or 22.2 mm aft of the front joiner. Leave battery travel available. This geometric quarter-MAC reference is not an approved flight CG range.

Balance in complete flight configuration. $CG\ movement = battery\ mass / total\ mass \times battery\ movement$. An experienced fixed-wing pilot must review stability, control authority and the final CG before flight.

Digital evidence

See the checks summary for recorded profile, mesh, foam and PDF-scale results. Record physical fit and load checks during the build.

Design basis: Onshape geometry, build transcript, Emax motor drawing and material specifications. CAD illustrations show part locations; colours distinguish materials.